🔑 Answer Key — Teacher Copy · Lesson 6-2

MSTAR PRACTICE WORKSHEET

Lesson 6-2: Division Expressions with Fractions and Mixed Numbers

Grade 6 Mathematics · Standard 6.NOS.1

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 SELECTED RESPONSE — CORRECT: C

What is the reciprocal of 3/5?

- (A) 1/3
- (B) 3/5
- (C) 5/3
- (D) 5/5

To find the reciprocal, flip the fraction. The reciprocal of 3/5 is 5/3.

Question 2 SELECTED RESPONSE — CORRECT: C

What is 1/2 ÷ 1/4?

- (A) 1/8
- (B) 1/2
- (C) 2
- (D) 4

1/2 ÷ 1/4 = 1/2 × 4/1 = 4/2 = 2. Two 1/4-size pieces fit into 1/2.

Question 3 SELECTED RESPONSE — CORRECT: B

What is 3/4 ÷ 1/8?

- A 3/32
- B 6
- C 8/3
- D 24

> 3/4 ÷ 1/8 = 3/4 × 8/1 = 24/4 = 6. Six 1/8-size pieces fit into 3/4.

Question 4 SELECTED RESPONSE — CORRECT: A

What is the first step in evaluating 2 1/2 ÷ 1/4?

- A Rewrite 2 1/2 as the improper fraction 5/2
- B Flip 2 1/2 to 1/2 2
- C Divide 2 by 1/4 and ignore the half
- D Add 2 + 1/2 + 1/4

> A mixed number cannot be flipped or multiplied as it stands, so rewrite it first: 2 1/2 = 5/2. Then Keep, Change, Flip gives 5/2 × 4/1 = 10.

Question 5 SELECTED RESPONSE — CORRECT: B

What is 2/3 ÷ 1/3?

- A 1/2
- B 2
- C 2/9
- D 6

> 2/3 ÷ 1/3 = 2/3 × 3/1 = 6/3 = 2. Two 1/3-size pieces fit into 2/3.

Question 6 SELECTED RESPONSE — CORRECT: C

A recipe needs 2/3 cup of sugar. You only have a 1/6-cup scoop. How many scoops do you need?

- (A) 2 scoops
- (B) 3 scoops
- (C) 4 scoops
- (D) 6 scoops

> 2/3 ÷ 1/6 = 2/3 × 6/1 = 12/3 = 4 scoops.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7 TWO-PART QUESTION

Part A — correct answer: C

What is the value of 2 1/4 ÷ 3/8?

- (A) 1/6
- (B) 2 2/3
- (C) 6
- (D) 27/32

> Rewrite 2 1/4 as the improper fraction 9/4. Keep 9/4, change ÷ to ×, flip 3/8 to 8/3: 9/4 × 8/3 = 72/12 = 6.

- **A:** That is the reciprocal of the answer. Keep 9/4 on top: 9/4 times 8/3.
- **B:** The wrong fraction was flipped. Keep 9/4 as it is and flip only the divisor, 3/8, to 8/3.
- **D:** That multiplies without flipping. Change 3/8 to 8/3 before multiplying.

Part B — correct answer: B

Which reasoning shows why your answer to Part A is correct?

- (A) Divide only the fraction parts, 1/4 ÷ 3/8 = 2/3, then put the whole number 2 back in front to get 2 2/3.
- (B) Change 2 1/4 to 9/4. Keep 9/4, change ÷ to ×, flip 3/8 to 8/3: 9/4 × 8/3 = 72/12 = 6.
- (C) Change 2 1/4 to 9/4, then multiply straight across without flipping: 9/4 × 3/8 = 27/32.
- (D) Change 2 1/4 to 9/4, then flip the first fraction instead of the second: 4/9 × 3/8 = 1/6.

> A mixed number has to become an improper fraction before keep, change, flip, and only the divisor gets flipped. That gives 9/4 × 8/3 = 72/12 = 6.

Question 8 WRITTEN RESPONSE · 2 POINTS

Type II Reasoning: Flipping the Wrong Fraction

Marisol divided 5/6 ÷ 2/3. Her work was: 6/5 × 2/3 = 12/15 = 4/5. She wrote that the quotient is 4/5.

Explain Marisol's misconception and give the correct quotient.

Model answer: Marisol flipped the dividend instead of the divisor. Keep the first fraction as it is, change ÷ to ×, and flip only the second fraction: 5/6 × 3/2 = 15/12 = 5/4 = 1 1/4.

2 points	Student explains that keep, change, flip flips only the divisor, not the dividend, so 5/6 stays as it is and 2/3 becomes 3/2, and gives the quotient 5/4 or 1 1/4.
1 point	Student gives the quotient 5/4 or 1 1/4 but does not explain that only the second fraction is flipped.
0 points	Student agrees with Marisol that the quotient is 4/5 or gives an unrelated response.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9 WRITTEN RESPONSE · 2 POINTS

Is 3/4 ÷ 1/2 the same as 1/2 ÷ 3/4? Solve both and explain why or why not. What does this tell you about the order in division?

Model answer: 3/4 ÷ 1/2 becomes 3/4 × 2/1 = 6/4 = 1 1/2, while 1/2 ÷ 3/4 becomes 1/2 × 4/3 = 4/6 = 2/3, so the two expressions are not equal. Swapping the dividend and the divisor flips the quotient to its reciprocal, which shows that division is not commutative — order matters.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.
